

Tianrun Gao



Date of birth

15 Aug 2000

China

Nationality

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Contact

Education

Sep. 2022 - Now

Tongji University, Shanghai, China

- Master Student of Civil Engineering
- Supervised by Prof. Hongwei Huang
- Anticipated Graduation: June 2026

Sep. 2018 - Jul. 2022

Tianjin University, Tianjin, China

- Bachelor of Engineering, Civil Engineering
- Ranking: 4/130, GPA: 3.86/4

Internship

Oct. 2025 – Now

Fudan University, Shanghai, China

- Internship of Artificial Intelligence Innovation and Incubation Institute and Shanghai Academy of AI for Science (SAIS)
- Supervised by Prof. Long Wei
- Research Focus: AI for PDE simulation/control, AI for fusion

Dec. 2024 – Oct. 2025

Westlake University, Hangzhou, China

- Internship of Department of AI, AI for Scientific Simulation & Discovery Lab
- Supervised by Prof. Tailin Wu
- Research Focus: Data-Driven Mechanics, AI for PDE simulation/control

Jul. 2021 - Nov. 2021

University of Toronto, Online due to Covid19

- Research Internship of Mechanical Reliability Analysis
- Funded by *Mitacs Globalink Program*, Canadian Government

Major professional courses

- Linear Algebra, Advanced Mathematics, Applied Statistics, Stochastic Process, Numerical Analysis, Theoretical Mechanics, Mechanics of Materials, Structural Mechanics and Dynamics, Elastic Mechanics and Finite Element Methods, Plastic Mechanics, Fluid Mechanics, Interdisciplinary Frontiers of AI and Physics, etc.

Research interests

- AI for physical simulation & design & control & discovery (especially on PDE & Fusion Energy).
- Mechanics and AI empowered mechanics.
- Robotics and Generative AI for science.
- Automated numerical simulation of mechanical & engineering systems and uncertainty analysis.

Computer and English Skills

- Software:
 - Proficient with Abaqus (FEM), FLAC (FDM), Python (Pytorch & Jax), Office, Origin, AutoCAD.
 - Familiar with Matlab, PFC^{3D} (DEM), MPM, Rhino.
- English Skills:
 - TOEFL: 102 (Reading:28, Listening:27, Speaking:23, Writing:24) (Nov. 2020).
 - GRE: 324 (Verbar Reasoning:155, Quantitative Reasoning:169, Analytical Writing:3.5)
 - Fluent in oral English, with fairly strong reading and writing ability.

Academic Contributions

Conferences (* represents equal contribution):

- [1] Peiyan Hu, Haodong Feng, Hongyuan Liu, Tongtong Yan, Wenhao Deng, **Tianrun Gao**, Rong Zheng, Haoren Zheng, Chenglei Yu, Chuanrui Wang, Kaiwen Li, Zhi-Ming Ma, Dezhi Zhou, Xingcai Lu, Dixia Fan, Tailin Wu. RealBench: A Benchmark for Complex Physical Systems with Real-World Data. ([Paper](#), [Project page](#), [Github repo](#).) (ICLR 2026, Oral.)
- [2] **Tianrun Gao***, Haoren Zheng*, Wenhao Deng*, Haodong Feng, Tao Zhang, Ruiqi Feng, Qianyi Chen, Tailin Wu. GenCP: Towards Generative Modeling Paradigm of Coupled physics with Application to Fluid-Structure Interaction. ([Paper](#), [Github Repo](#).) (ICLR 2026, Poster.)
- [3] Tian Xia, **Tianrun Gao**, Wenhao Deng, Long Wei, Xiaowei Qian, Jiang Yixian, Chenglei Yu, Tailin Wu. BuildArena: A Physics-Aligned Interactive Benchmark of LLMs for Engineering Construction. ([Paper](#), [Project page](#), [Github repo](#), [Demo](#).) (Submitted to ICML 2026.)
- [4] Qianyi Chen*, **Tianrun Gao***, Chenbo Jiang*, Tailin Wu. EqCollide: Equivariant and Collision-Aware Deformable Objects Neural Simulator. ([Paper](#).) (Submitted to KDD 2026.)
- [5] **Tianrun Gao**, Dongming Zhang, Xiaomin Liu, Hongwei Huang. Data-based Risk Evaluation on 4 Overlapped Existing Subway Tunnels Undercrossed by Shield Tunneling. ITA WTC 2025, Stockholm, Sweden, 2025.
- [6] **Tianrun Gao**, Dongming Zhang, Hongwei Huang. Experimental Study on Tunneling-Induced Disturbance and Propagation in Complex Environment. 2025 PIARC, Chongqing, China, 2025.
- [7] **Tianrun Gao**, Dongming Zhang, Hongwei Huang. Study on Predicting Existing Tunnel Settlement Induced by Shield Tunneling Based on Machine Learning. The 2nd Workshop on Future of Machine Learning in Geotechnics and the 5th Machine Learning in Geotechnics Dialogue, Chengdu, China, 2024.
- [8] **Tianrun Gao**, Jianwei Jia, Xiaomin Liu, Wenjun Zhang, Hongwei Huang. 3D refined numerical simulation analysis of the impact of shield tunnel construction on adjacent pile foundation and soil. 2022 China Tunnel and Underground Engineering Conference (CTUC2022), Changsha, China, 2023.

Publications:

- [1] Hongwei Huang, **Tianrun Gao**, Dongming Zhang. A Hybrid Approach for Modifying Tunneling-Induced Response in Existing Multi-Tunnel Environment. Computers and Geotechnics, 2025, 179, 106921.
- [2] Jianzhong Zhang, **Tianrun Gao**. Compressibility of Abnormal Pressure Gas Reservoirs and its Effect on Reserves. ACS omega, 2021, 6(40): 26221-26230.
- [3] Rui Jia, **Tianrun Gao**, Gang Yang. Analysis on the Influence of Construction Stress Release and Disturbance on Seismic Response of Tunnel. Chinese Journal of Underground Space and Engineering, 2022, 18(S2): 916-925.

Patents:

- [1] Hongwei Huang, Jinzhang Zhang, **Tianrun Gao**, Dongming Zhang. Experimental Model for Full Cross-Section Deformation Monitoring of Longitudinal and Transverse Tunnels and Post-Processing Algorithm Based on GANs. Shanghai, China, 2024-10-31. (*Invention Patent Officially Accepted for Review*)
- [2] Zhixing Chang, Xuesong Cheng, Hongfei Cheng, **Tianrun Gao**, D. Y. Li, R. Z. Wang. A 3D Printing Device for Underground Pipeline Structure Installation. Tianjin, China: CN20212 0691496.7, 2021-11-05. (*Granted*)
- [3] Zhixing Chang, Xuesong Cheng, Hongfei Cheng, **Tianrun Gao**, D. Y. Li, R. Z. Wang. A 3D Printing Device for Underground Pipeline Structure Installation. Tianjin, China: CN202110 367402.5, 2021-07-09. (*Granted*)

Other information

- Self-motivated, with a passion for learning and the capability to efficiently master new knowledge.
- Physically fit and passionate about sports, served as a member of Tianjin University Rowing Team winning 1st prize in Tianjin, demonstrating strong sense of responsibility and team spirit.
- Quick-minded with strong communication skills, achieved the University Debate Championship as a member of debate team.
- Easy-going, optimistic, adaptable and resilient.